

第一部分
无线QoS业务监测

TMO18416_V6.0-SG-EN-(T)LA6.0-Ed1 Module 1.1 Edition 6.0

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单元目标



通过本单元的学习，你应该能:

- 描述用于LTE的服务质量监测工具
- 描述 NPO的作用和它如何与 SAM(操作和维护)及网络单元进行相互作用，比如 eNB和 MME (移动管理实体)
- 列举计数器和指标的不同
- 解释触发器什么时候和如何被触发
- 识别不同类型的指标，了解它们是怎样工作的.

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3GPP Counter Summary (Common / FDD / TDD)

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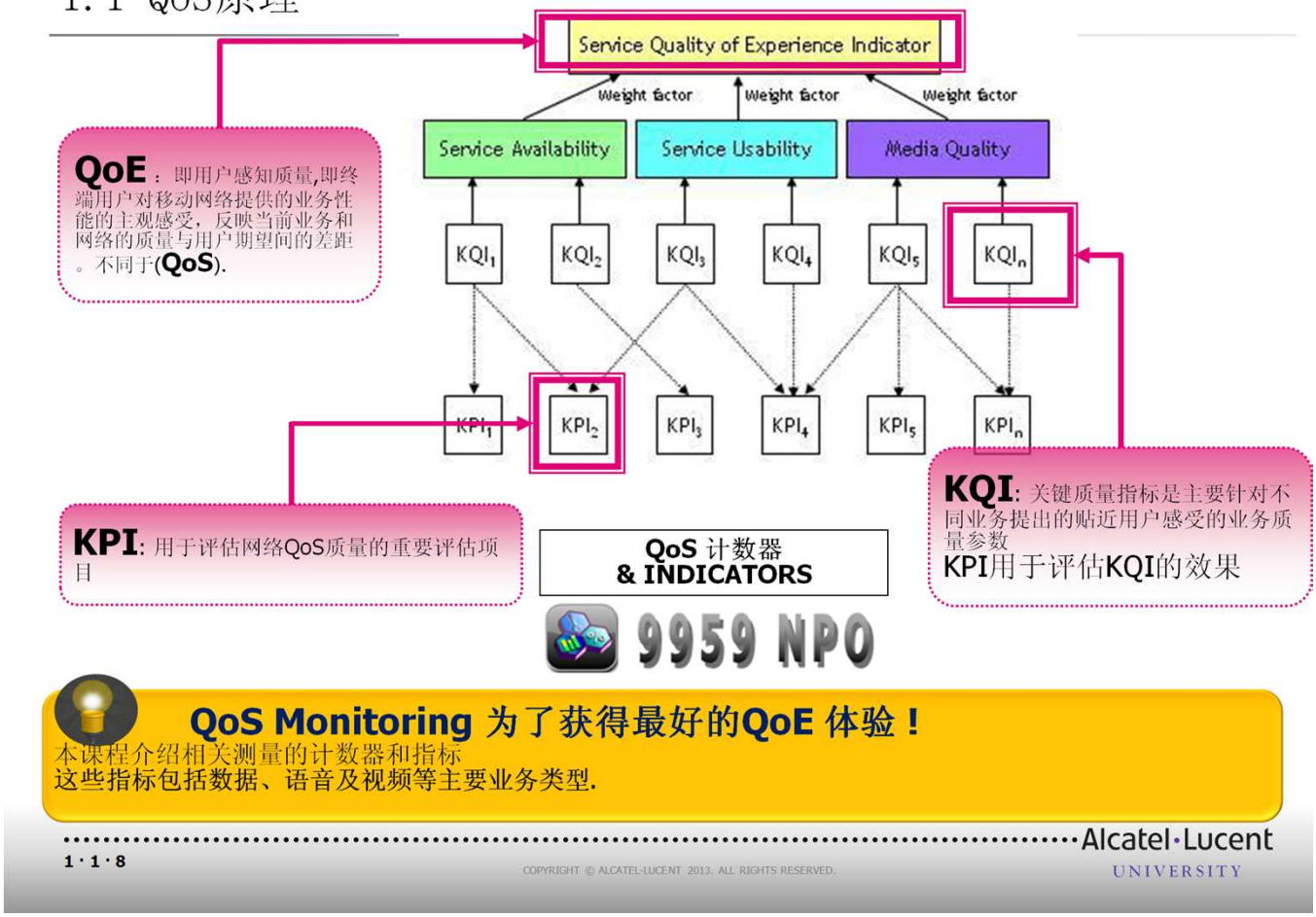
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1 QoS监测工具介绍

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1.1 QoS原理



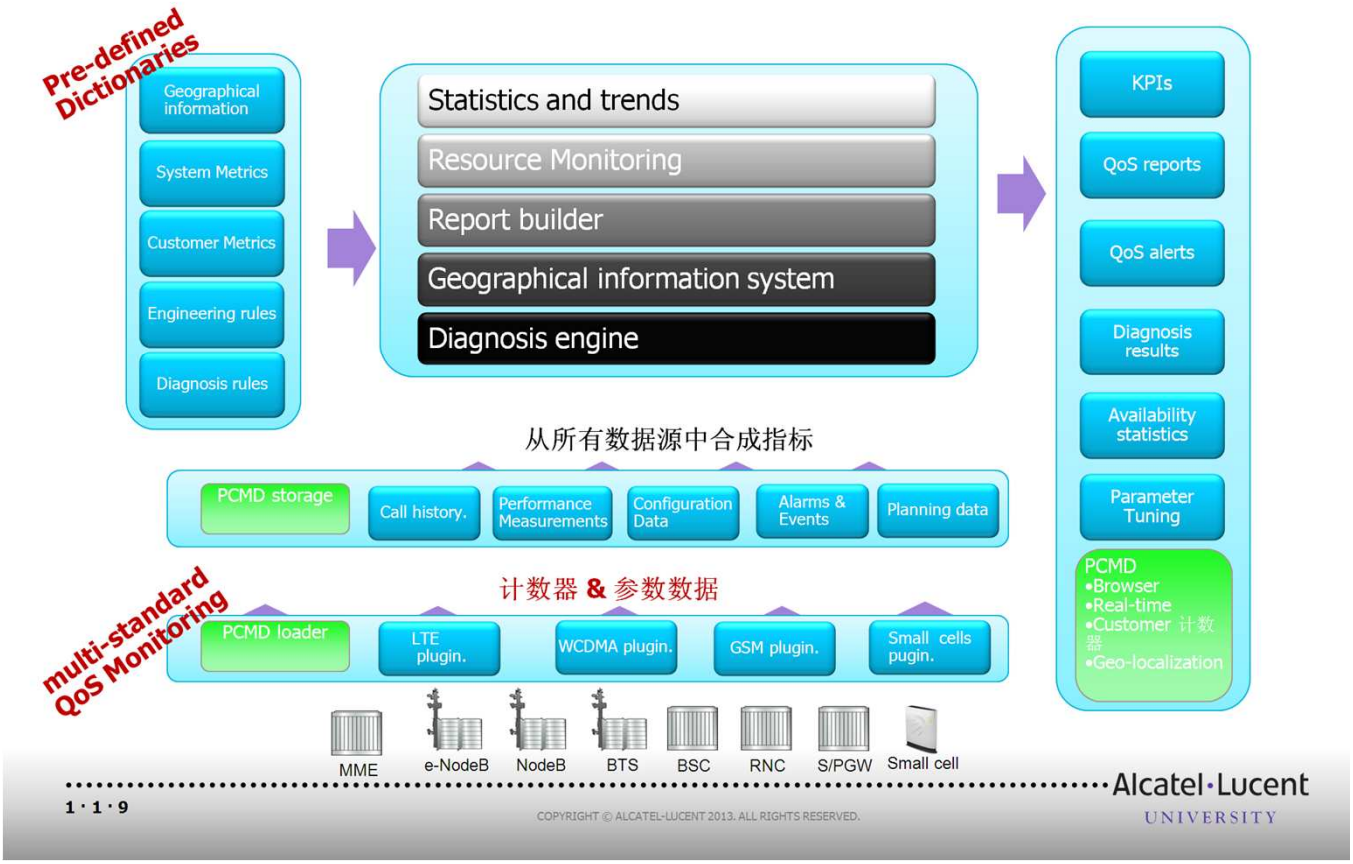
NPO 使用QoS计数器构建ALU的初始指标及定制指标

QoE 和KQI不在本课程中介绍。

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1.2 网络性能优化工具

9959 NPO



NPO 能够提供多系统GSM/WCDMA/LTE的优化支持



NPO add-on are available to refine QoS analysis

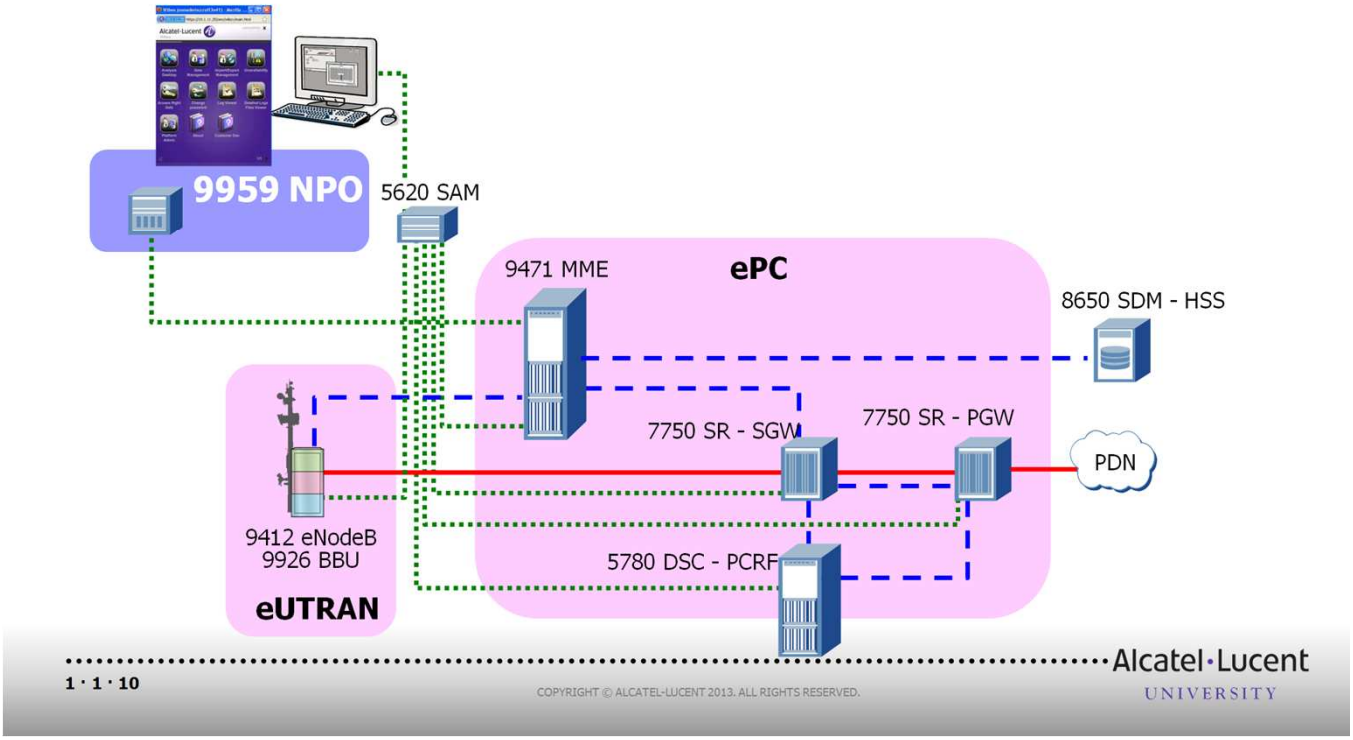
✓**NPO-PCMD** includes LTE PCMD flexible counter editor allowing counter design at OAM level. NPO PCMD counter are computed at NPO level. It is possible to create a new scalar counter on field that can be exported/imported (to allow sharing counter definitions across teams).

✓**NPO-NUART** (Network Unavailability Analysis and Reporting Tool) for End To End global performance analysis.

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1.3 网络性能优化工具

- **用集中的方式监测网络的服务质量**
- **把所有的QoS数据和QoS服务集成到一个应用中**



这些计数器可以作为计算度量值的基准(一个度量值仅仅能在一台给定的电脑上使用).

- 对于每个计数器, 它的产品说明包含:
- 计数器属性 (3GPP 名字, 计数器类型, 位置...),
- 计数器定义 (触发时刻),
- DFD采样来强调话务统计 (DFD在某些情况下被来自同一家庭的所有计数器共享).
- 计数器通过族集合.

在今后，一些计数器将重复地使用一种方式定义: 筛选过的和没有被筛选过的相似的计数器都会提供. 这就必须有: 度量计算要使用的综合信息, 详细描述将要使用的信息比如故障排查.

另外, 要避免无用的计算来获得例如监测事件的总数，使管理那些引入额外耗费的情况成为可能 (即使筛选错误的时候，总数也应该正确).

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1.3 LTE的 NPO功能

QoS分析

- **采集测量值 （可保存两年）**
- **生成指标（超过10,000个）**
- **提供一个强大的动态图形用户界面可以让 QoS报告分析变得更容易**
- **一些特殊功能可以鉴别出与 QoS指标有关的最好或者最坏的单元**

参数的参考值

规则和分析

趋势预测

配置参数



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2 计数器

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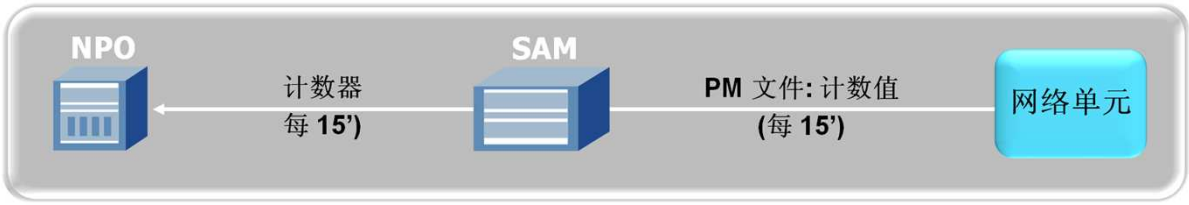
2.1 计数器定义

计数器是由 eNodeB产生的, 它们检测在给定的一段时间内的一些特定事件的发生 (发生在 eUTRAN的实体)

eNodeB 提供一个 PM文件，包含了在叫做“粒度周期”的既定时间内的计数器的值，并且每5′, 10′ 或 15′变化一次.

然后计数器的值储存在 SAM服务器中

NPO 收集该 PM文件



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文档中使用的措辞计数器和测量术语是为了区别同一个概念. 通常, Alcatel-Lucent使用术语计数器来区别来自（无线）测量结果的计数器. 计数器以分钟或小时为单位周期性地描述测量结果. 例如, 15分钟.

当一个测量结果以500 ms为周期来描述测量术语.

3GPP性能管理规范更适宜使用测量结果. 当环境是 3GPP规格时, 使用测量术语, 尽管在 Alcatel-Lucent特殊部分使用计数器, 所有的措辞都是等价的.

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2.1.1 计数器属性

Counter Information	Counter Value/Description
Counter Code	13204
Counter Type	LOAD
Triggering (Event)	This counter is triggered when a VoIP bearer is setup or released in the cell (successful or with partial failures procedures, EPC-initiated procedures, incoming and outgoing mobilty procedures and generally all procedures involved in E-RABs setup and E-RABs release may trigger this counter when performing the CAC).
Subcounters	Not defined
Subfamily	Bearers
3GPP name	VS.NbVoIPBearersPerCell
Object Instance	EutranCell
Range	0 to 2^{31}-1
Unit	Bearer

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●计数器名字:这部分包括计数器名字.

●计数器编号:这部分包括计数器编号.

●计数器类型:这部分包括计数器字段,它用来在结果文件中确定计数器的类型比如累加的，负载的或者数值的.

●触发事件:这部分包括引起将要更新的测量结果的条件: 该条件是通过确定和触发事件有关的协议来定义的，这些事件要么会启动或停止测量进程，要么更新当前测量结果.如果不能提供精确条件,有条件的话将会定期更新.

●如果 Alcatel-Lucent实现符合 3GPP规格，那么声明就是 "3GPP specified counter".

●如果 3GPP规格没有考虑到计数器，那么声明就是"Alcatel-Lucent specified counter".

●对象实例: 这部分包括对象类别和实例. measObjInstId 字段确定测量对象类别和它的实例.

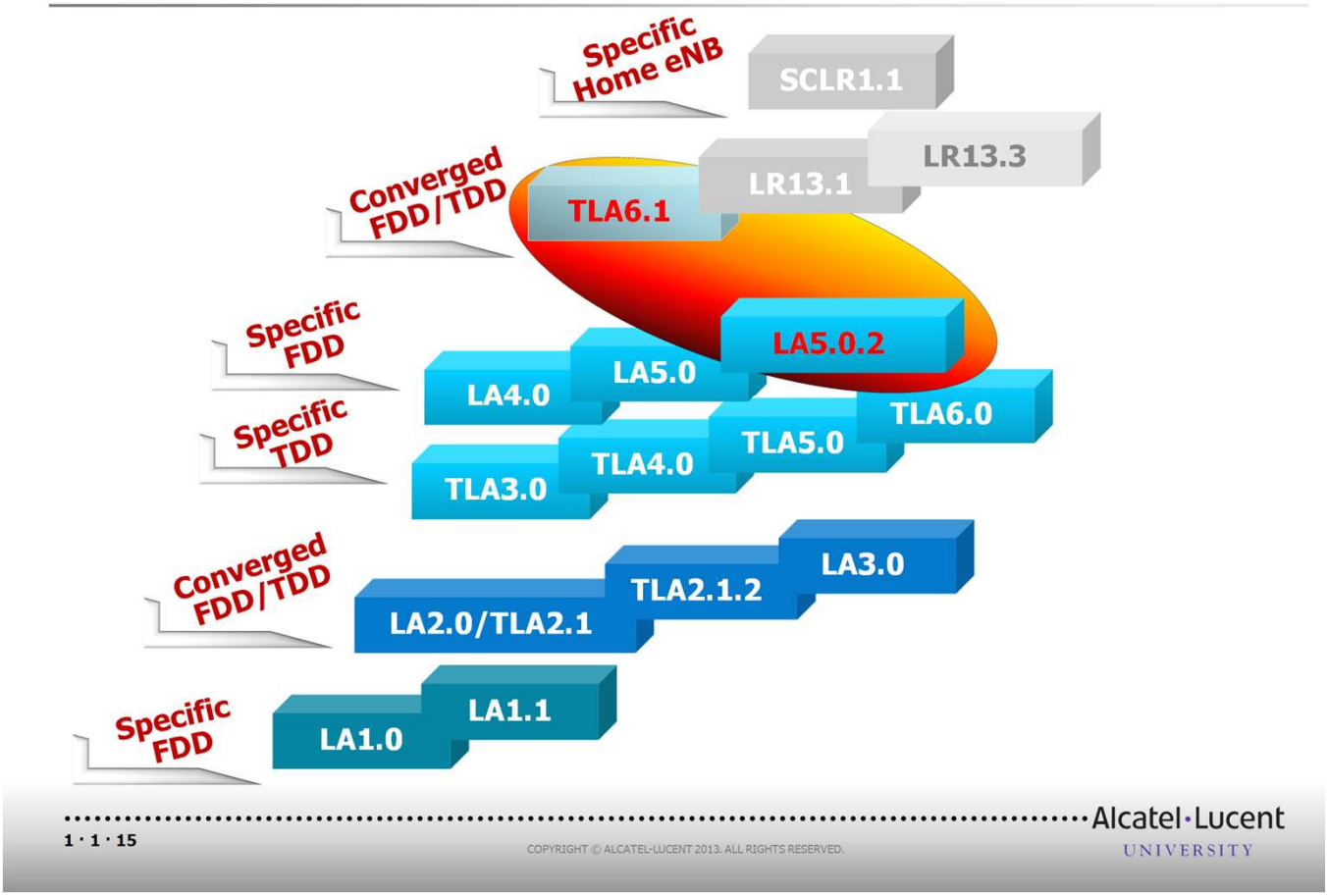
●范围: 这部分包括计数器范围.

●单元: 这部分包括计数器单元.

●可用信息：这部分阐述了 Alcatel-Lucent LTE RAN系统在引入计数器的情况下的可用性.

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2.1.1 eNodeB PM 计数器字典版本



TDD LTE与FDD LTE有少量的计数器不一样

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2.1.2 计数器聚合规则

Rules	Results	Type
Rtotal	X.cum = X1.cum + X2.cum +... + Xn.cum	CUM (CC) 计数器
Rload/Rval	X.cum = X1.cum + ... +Xn.cum X.nbevt = X1.nbevt + ... +Xn.nbevt X.avg = (X1.cum + ... +Xn.cum) / (X1.nbevt + ... +Xn.nbevt) X.min = min (X1.min,...,Xn.min) X.max = max (X1.max,...,Xn.max)	Some LOAD and VAL 计数器
Rload*	Same as Rload but resulting X.cum has no physical meaning due to the nature of meaning due to the nature of the measurement (like percents) and is not interpreted but is necessary for intermediate computations.	Some LOAD 计数器

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每个 GP, NPO 将会处理从 O&M服务器 (SAM)获得的 PM XML文件

NPO有计数器的映射 (和关联的参考身份)，它会在对应的消息中的 PM XML 文件显示.

NPO 不支持 ".” 所以对于所有的计数器名称 the ".” 自动被 "_”所替换. NPO 计数器 将会显示 "_” 而不是 ".”

在 NPO AD (分析桌面)中, 对与对应的对象可以显示计数器. 为此, 你必须选择对象和 “计数器” 的标签 为了更好的理解，显示通过 族/子系列 (在计数器定义中规定)

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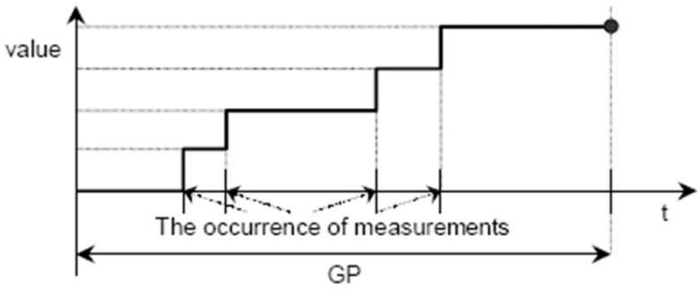
2.1.3 累加计数器

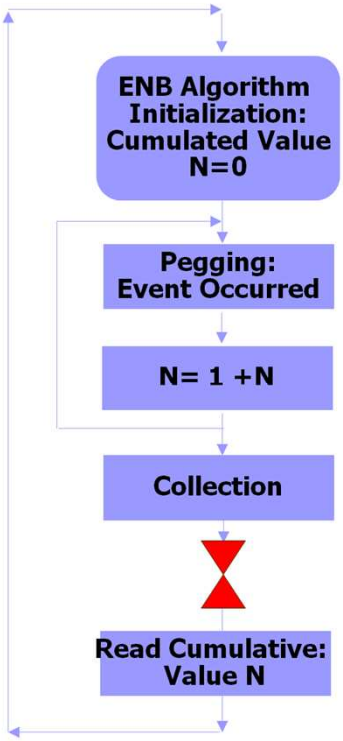
累加计数器是最简单的类型.

在粒度周期内，累加计数器通过检测既定事件的捕获数据.

累加计数器所提供的测量数据的方式与实现定义的事件有关.

结果是单个字段,"计数值".





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- 累加计数器提供原始的计数. 累加计数器是 CUM(CC)类型的计数器, 在 3GPP TS 32.104规定中有说明. 当计数事件发生时，计数值将会自加,它能提供累加值. NE累加计数器在 OAM GUI层面上提供计数值. 累加计数器的类型是最简单的.
- 在粒度周期内，累加计数器通过检测既定事件的捕获数据.累加计数器所提供的测量数据的方式与实现定义的事件有关. 结果是单个字段,"计数值".

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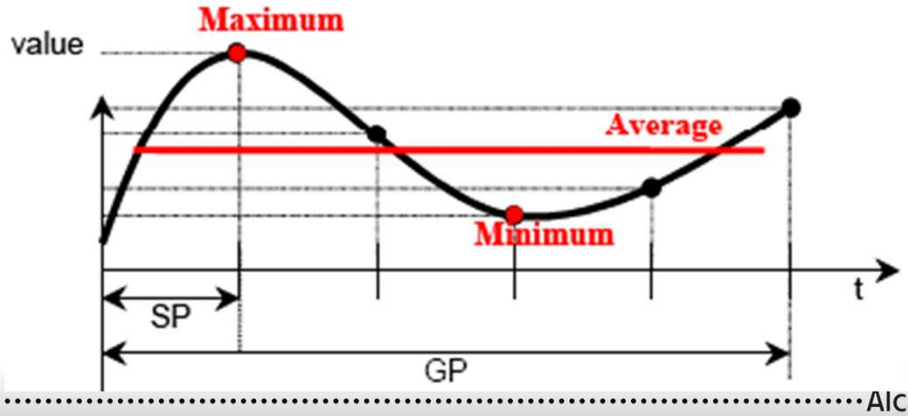
2.1.4 值计数器

在粒度周期内，累加计数器通过检测既定事件的捕获数据.

这就是说, 每一次事件发生都会提取一个测量数据; 然后在对于的粒度周期内，每个测量数据都将会加到累加计数器中.

值计数器被称为 DER – 离散事件计数器.

值计数器可以提供最小值, 最大值, 事件的数量和累计值.



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值计数器也叫离散事件寄存器 (DER). 值计数器提供最小，最大和平均值.

每次事件提供一个计数值，它都会加上累加值. 平均值是累加值除以事件发生的次数，eNodeB不会报告它，但是在 NPO层面上会计算它.

在粒度周期内，累加计数器通过检测既定事件的捕获数据.每一次事件发生都会提取一个测量数据; 然后在对于的粒度周期内，每个测量数据都将会加到累加计数器中.

在粒度周期内，每个值计数器都会报告以下的计数值:

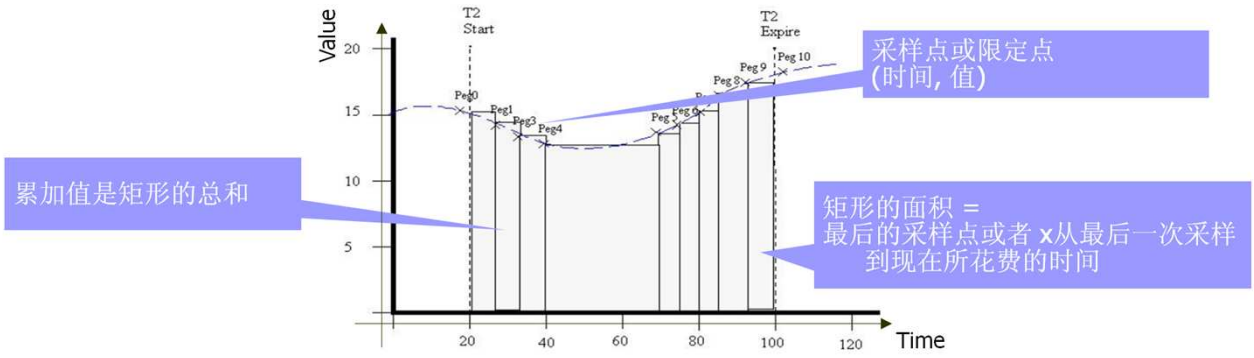
- 累加值 (.Cum in XML files)
- 事件发生的次数 (.NbEvt in XML files)
- 最大值 (.Max in XML files)
- 最小值 (.Min in XML files)

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2.1.5 负载计数器

有2种类型的负载计数器:

- **周期采样的负载计数器 (SI – 状态检测): 它**源于一个特定测量的采样, 在粒度周期内每隔一段时间进行一次.
- **在既定事件发生时进行采样的负载计数器:** 特定测量可能也需要，不必相同时间间隔, 但是在粒度周期内，由既定事件的发生与否来触发采样.



NbEvent = Total Elapsed Time (donc = GP)

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有2种类型的负载计数器:

1. 周期采样的负载计数器 (SI – 状态检测),源于一个特定测量的采样, 在粒度周期内每隔一段时间进行一次. 这就是说, 在对应的粒度周期内，每个采样结果都会加到累加计数器中. 累加计数器用来分析平均值，趋势，或者循环. 周期采样的负载计数器也会进行状态检测. 周期采样的负载计数器提供最小值，最大值和平均值. 计数器的类型是以在一个限定的速度下的自激理论为基础的. 该计数器提供能迅速捕获变化数据的方式.

2. 在既定事件发生时进行采样的负载计数器, 它完成特定测量时不在相同的时间间隔采样，但是在粒度周期内，由既定事件的发生与否来触发采样.

在粒度周期内，每个采样值都会加到累加计数器中. 累加计数器用来分析平均值，趋势，或者循环. 这是第2种负载计数器，但是它没有标准中　提到的状态检测. 然而, 这两种类型的负载计数器, 记录相同的数值，并且使用相同的算法来计算下列数值:

Alcatel-Lucent提供以下数值:

• 累加值 (.Cum in XML files), • 采样的数量 (.NbEvt in XML files)

• 最大的采样值 (.Max in XML files), • 最小的采样值 (.Min in XML files)

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2.1.6 计数器的参考 ID类型

计数器可以是:

- **一个简单的计数器:**

参考标示符是 LCXyyyyy.

例: VS.RrcConnectionSuccessSum, VS.RrcConnectionFailureSum.

- **一个筛选计数器:**

参考标示符是 LCXyyyyy_zz.

它是“根”计数器的具体原因的一种. 看到这种类型的参考标示符就能知道它有若干原因 .

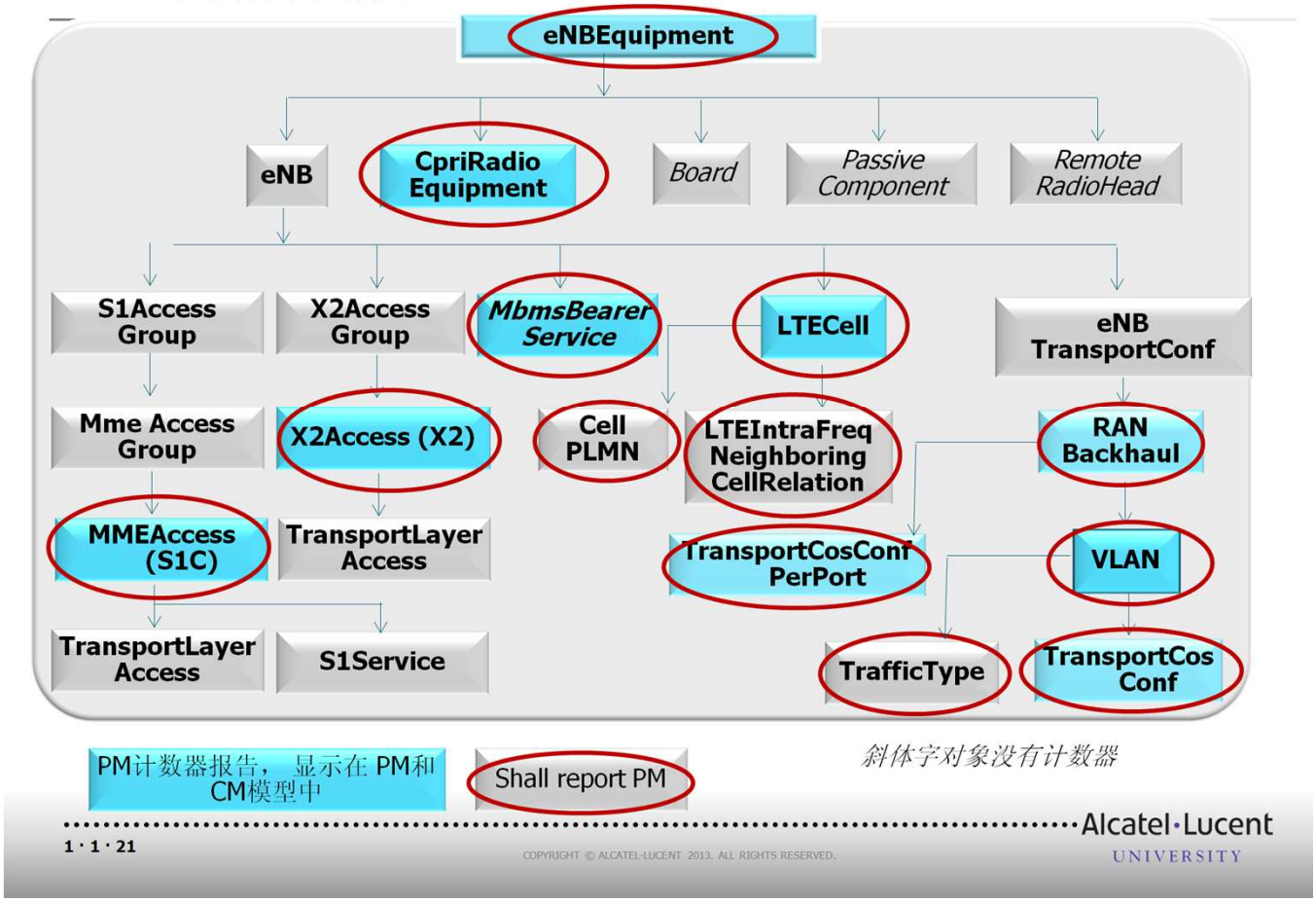
例:

VS.RrcConnectionRequest.EmergencyCallAttempts,VS.RrcConnectionRequest.HighPriorityAccessAttempts, VS.RrcConnectionFailure.CACFailure, VS.RrcConnectionFailure.NoResponseFromUE.

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2.2 对象层次结构-LA6.0



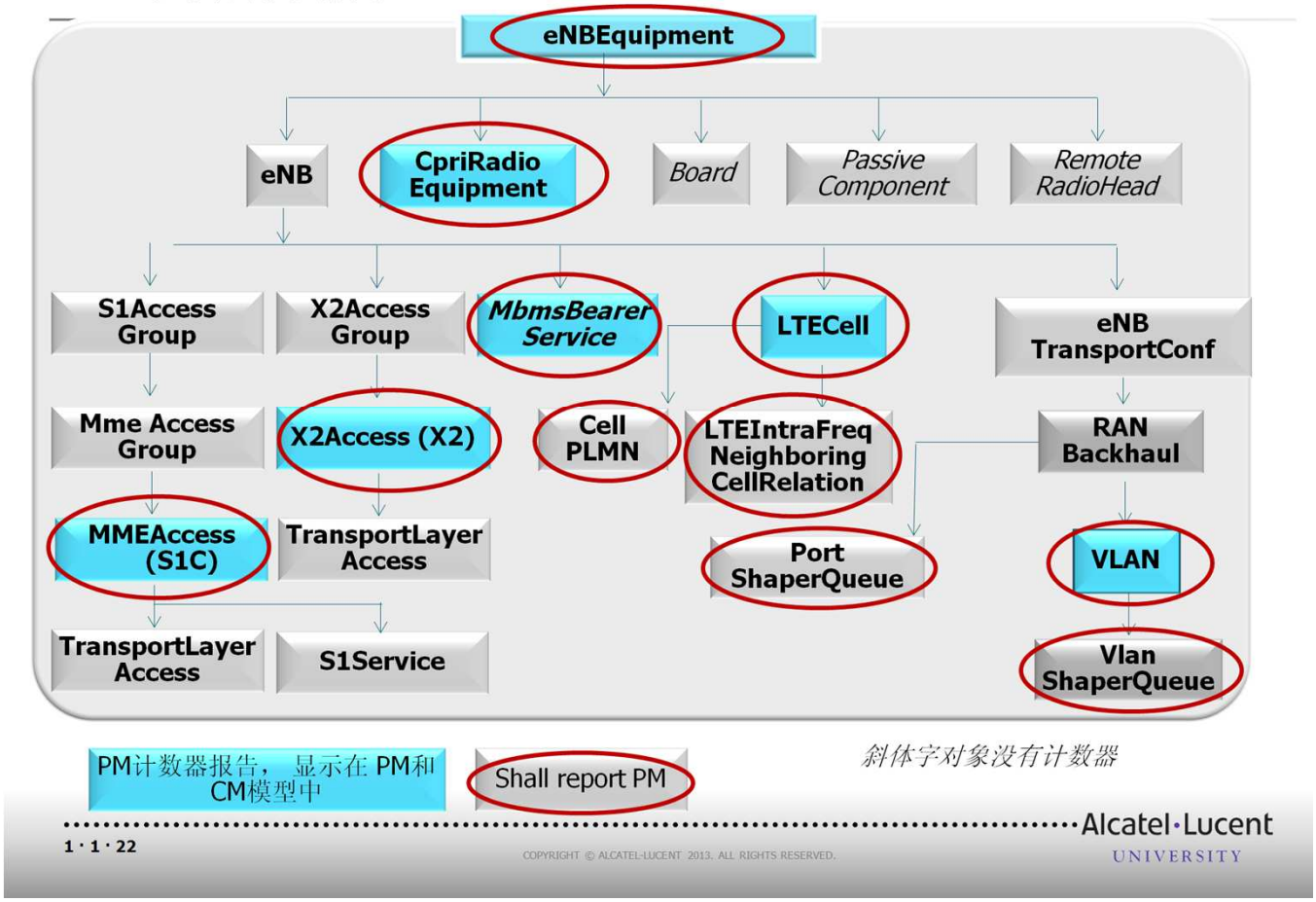
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计数器关于对象的报告在层次结构中一一阐述. 资源对象是指 eNodeB的本地管理信息模型(MIM)的管理对象的分支. 这个模型在业务经理(SAM)与 eNodeB的接口中使用. 在检测对象模型的性能中, 每个对象类别有该类别所支持的计数器的一个独特的集合. 它展现了 eNodeB MIM的一个分支. 这幅图片描述了 eUTRAN的一个潜在的未来测量对象的树结构.

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2.3对象层次结构-TLA6.1



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2.1.8 3GPP计数器族



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文档中的计数器被分为计数器族. 计数器族附属于既定的LTE功能群. 例如, UE环境管理族把所有的计数器关联到UE环境管理(建立, 修改, 检测,

释放).

Alcatel-Lucent 所定义的计数器族不同于 3GPP TS 32.425所定义的 3GPP测量族.

Alcatel-Lucent对于计数器的规定, 计数器命名 (在下面的部分也叫 “3GPP Name”(p. 1-11)) 以前缀 “VS”开始. 例如, VS.IncomingInterENodeBS1HOFailure.CACFailure.

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3 指标

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3 指标定义

- NPO通过处理 PM XML文件来提取计数器的值 (也称作原始数据).
- 储存计数器的值是在指标中完成的.
- 一个指标指定了一个名词和参考标示符.
- 从 NPO的角度来看，有2中类型的指标:

  - 基本 (也叫存储) 指标:

  对于这些指标, NPO 存储了数据库中的值. 每当用户拖&放这种类型的指标, NPO 将会读取数据库中的值。

  - 计算指标:

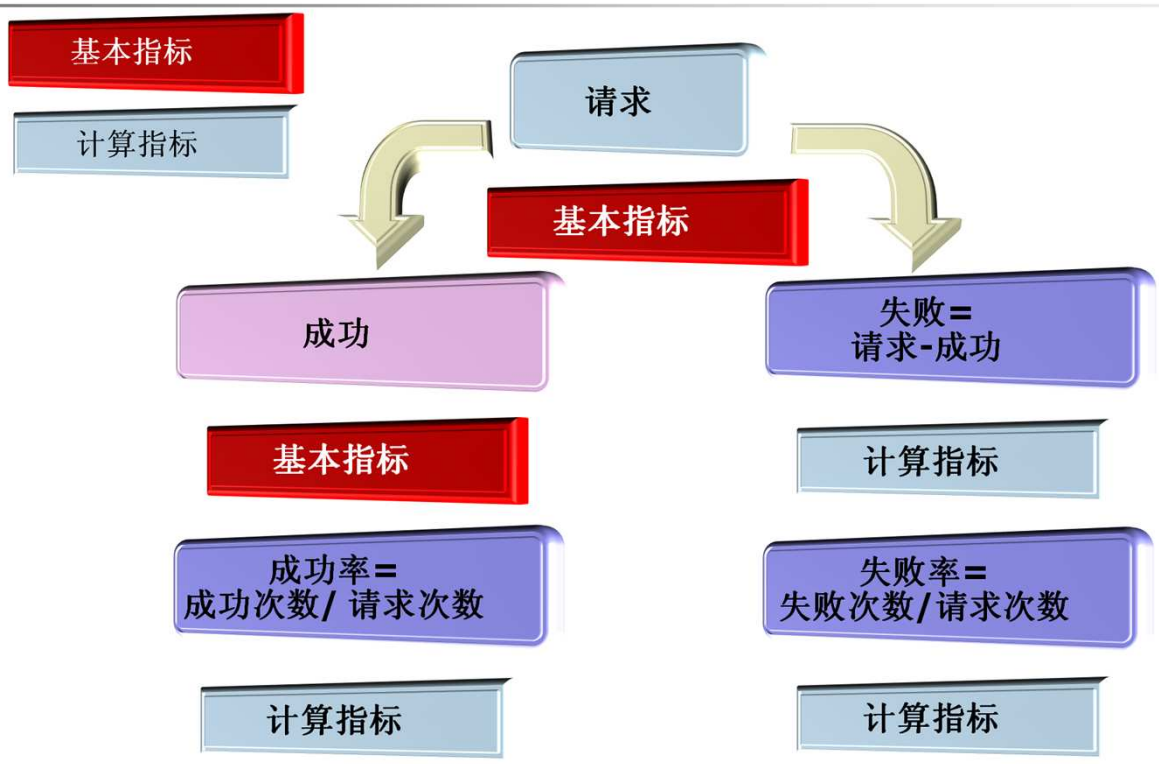
  这些指标进行 "现场" (从不存储)计算.

  每当用户拖&放这种类型的指标, NPO 以基本计数器为基础来计算数值，然后把它显示。

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3 指标定义 [续]



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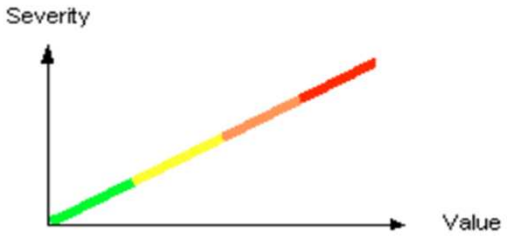
3.1 指标属性

Parameter	Description
Reference Name	The internal short identifier for the indicator.
Long Name	The external identifier for the indicator.
Description	The description of the indicator and the associated subindicators.
Family	The family to which the indicator belongs.
SubFamily	The subfamily to which the indicator belongs.
Family1	The family1 to which the indicator belongs.
SubFamily1	The subfamily1 to which the indicator belongs.
Formula	The formula to calculate the indicator.
Unit	The indicator value measurement unit. It can be packets, kBs, event, or percentage.
Trend Supported	The ability to see trends of indicators following the day of week or the hour of day to predict the behavior of the network.
UpperIsFault	The flag that indicates the quality of indicator. If this flag is set to true, it indicates issues in the network.

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3.1.1 指标范例: VS_UE_ctxt_modif_eNB_setup_fail

Item	According to
Reference Name	L12511_20_CI
Long Name	VS_UE_ctxt_modif_eNB_setup_fail
Description	This indicator provides the number of UE context modification failure. NOTES: The counter and hence the indicator is specific to TDD.
Family	QoS (global indicators)
SubFamily	Bearer session management
Family1	QoS (global indicators)
SubFamily1	UE context management
Indicator Formula	VS_UE_ctxt_modif_eNB_setup_req - VS_UE_ctxt_modif_eNB_setup_succ
Unit	No unit
Trend supported	True
Upper is fault	

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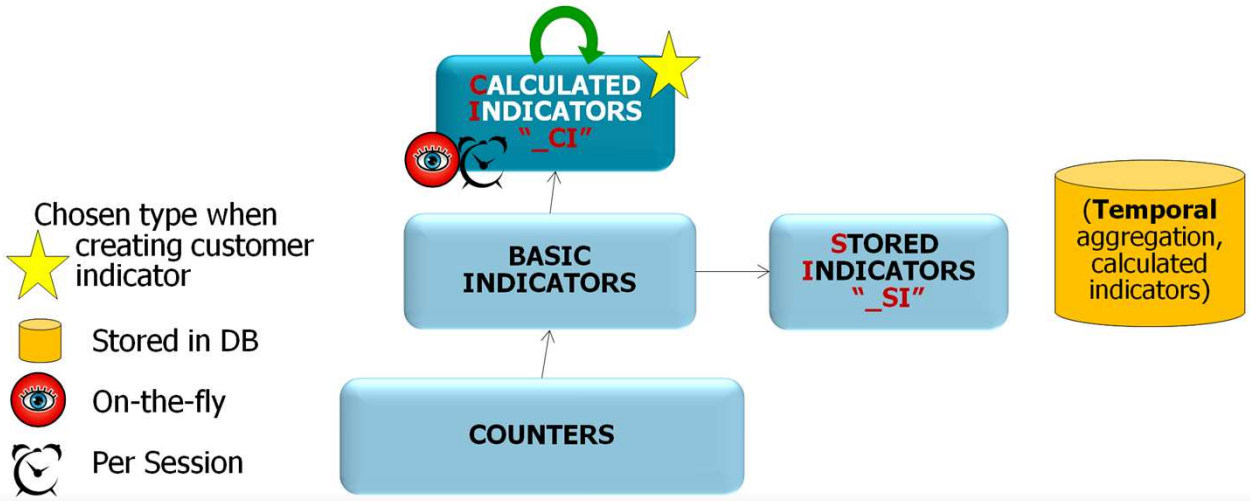
3.1.2 NPO 计数器和指标总结

NPO 计数器名称：LC12311

('LC' 即'LTE Counter'
来自 1 eNodeB node)

NPO 指标可能是:

- 1个计数器构成的 Basic Indicator *(LC12311= L12311)* (静态存储类型)
- 由Basic Indicator通过算式计算得到的统计指标



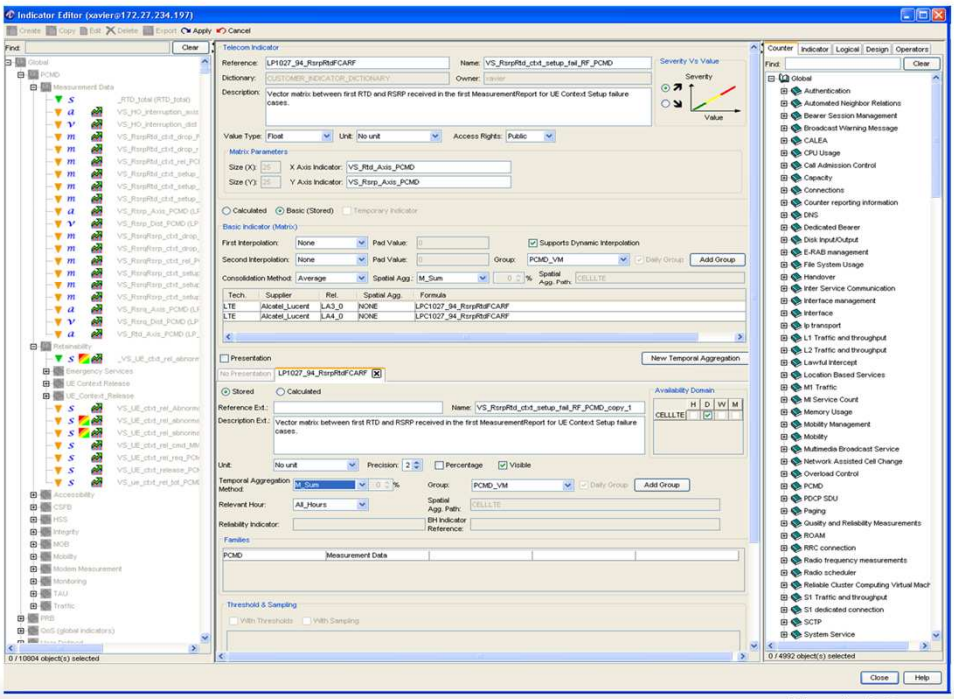
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NPO可以基于数据库保存的基础指标，定制需要的指标，保存在用户会话中。

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3.1.3 多版本共用指标

NPO 字典中预定义的Multi-release INDICATORS



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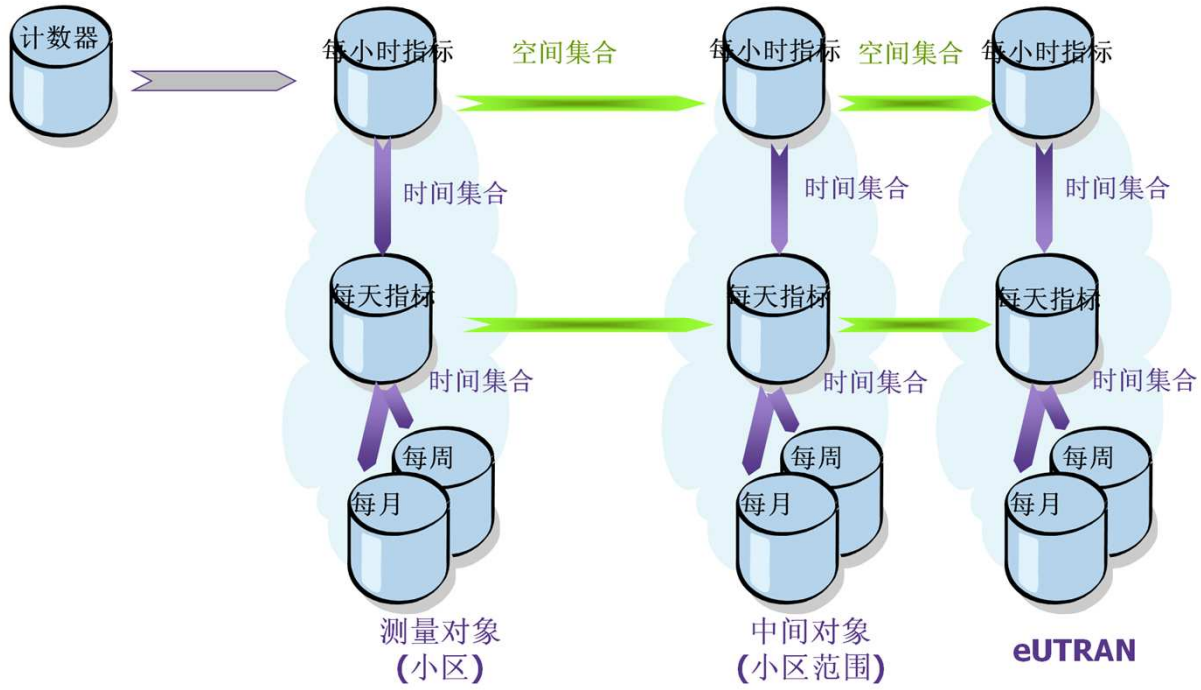
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这些指标在多个版本下均能正常工作，KPI建议使用这些指标定义

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3.2 合并

- 指标是合并生成的. 合并通过集合构成 (空间或时间的).



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4 课程主要模块简介

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4.1 LTE 计数器家族和标识范围

LTE eNodeB Counter 范围由下表定义

接下来本课程按照计数器的"Monitoring groups"来介绍:

2. Service Accessibility
3. Retainability
4. Mobility
5. Traffic
6. Quality

Counter family	Counters id range
RRC connection	12302 .. 12321 (19 counters)
S1 dedicated connection	12401 .. 12405 (4 counters)
UE context management	12501 .. 12513 (13 counters)
E-RAB management	12603 .. 12637 (35 counters)
Mobility	12701 .. 12898 (178 counters)
PDCP SDU	14201 .. 14206 (6 counters)
Miscellaneous	14701 .. 14701 (1 counters)

	Counter family	Counters id range
	L1 Traffic and throughput	12001 .. 12068 (60 counters)
	L2 Traffic and throughput	12101 .. 12129 (22 counters)
	Counter reporting information	12200 .. 12202 (3 counters)
	X2 Traffic and throughput	12909 .. 12912 (4 counters)
	Radio scheduler	13001 .. 13019 (19 counters)
	S1 Traffic and throughput	13109 .. 13112 (4 counters)
	Capacity	13201 .. 13249 (47 counters)
	Ip transport	13301 .. 13342 (38 counters)
(*)	eNodeB synchronization	13401 .. 13410 (10 counters)
	Paging	13501 .. 13502 (2 counters)
	SCTP	13601 .. 13610 (10 counters)
	Radio frequency measurements	13701 .. 13705 (5 counters)
	Call admission control	13801 .. 13808 (8 counters)
(*)	S1-c Ippa Traffic	13901 .. 13906 (6 counters)
	Interface management	14101 .. 14102 (2 counters)
	M1 Traffic	14301 .. 14316 (16 counters)
	UE radio parameter management	14401 .. 14402 (2 counters)
(*)	eMBMS signaling	14501 .. 14520 (20 counters)

1 · 1 · 33

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本课程不涉及PCMD的计数器介绍。

On NPO,

One Indicator can be associate in two different family /subFamily/subsub...
It allows to define filtering groups, such as main indicators group for customer KPI purpose or special traffic services follow up.

***KPI** family example:*

- *FAMILY: **QoS** (global indicators)*
- *SUBFAMILY: **Inter Operability***
- *SUBSUBFAMILY: **Network performance***

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4.2 KPI相关QoS指标家族



QoS优化常用的指标家族如下

以下属于*NPO "subFamily1"*

- ***QoS/RRC Connection***
- ***QoS/S1 Dedicated Connection***
- ***QoS/Bearer session management***
- ***QoS/Capacity***
- ***QoS/E-RAB management (all, per PLMN, per VLAN)***
- ***QoS/Handover***
- ***QoS/L2 Traffic and Throughput***
- ***QoS/Mobility management***
- ***QoS/UE context management***
- ***QoS/Inter Operability (*)***



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KPI定义需要根据客户需要选择相应指标和算式

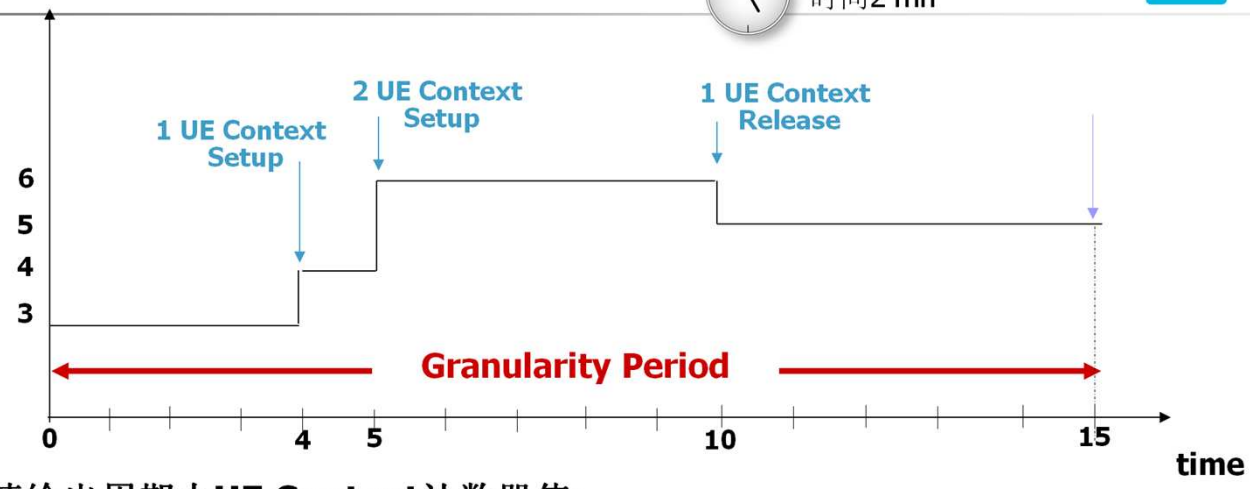
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5 练习

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5.1 计数器统计

时间2 mn



请给出周期内**UE Context**计数器值:

#.Min =

#.Max =

#.Cum =

#.NbEvt =

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#.NbEvt指周期内发生的事件总次数

#.Cum 即全部事件的Context累加值

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5.2 临时指标的合成

时间1 mn

临时指标的时间单位序列是:

1. Hourly / Weekly/ Monthly / Yearly periods ☐
2. Hourly / Daily / Weekly / Monthly periods ☐
3. Hourly / Daily / Monthly / Yearly periods ☐

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5.3 回顾：UE测量的内容?



时间5 mn

LTE measurements from UE:

UMTS FDD/TDD measurements from UE :

GSM measurements from UE :

UE CDMA2000 HRPD measurements from UE :

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空闲模式下的测量项目。

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3GPP Counter Summary (Common / FDD / TDD)

id	counter name3GPP
2001	VS.VoIPDLFER
2003	VS.CellDLL1Throughput
2004	VS.CellULL1Throughput
2007	VS.DLResidualMacBLERWithDynamicScheduling
2008	VS.DLInitialMacBLERWithDynamicScheduling
2009	VS.ULResidualMacBLERWithDynamicScheduling
2010	VS.ULInitialMacBLERWithDynamicScheduling
2011	VS.DLDataVolumeWithDynamicSchedulingPerUserCategory
2013	VS.ULDataVolumeWithDynamicSchedulingPerUserCategory
2015	VS.DLPRBUsedWithDynamicSchedulingPerUserCategory
2017	VS.ULPRBUsedWithDynamicSchedulingPerUserCategory
2019	VS.PUCCHMessagesPerType
2023	VS.PUCCHCQIPeriodHistogram
2024	VS.PUCCHSRPeriodHistogram
2025	VS.PUCCHSRSPeriodHistogram
2026	VS.CFIUsage
2027	VS.ULNoisePerPRBGroup
2028	VS.ULTotalPRBUsage
2029	VS.DLTotalPRBUsage
2030	VS.CellDLL1ThroughputLoad
2031	VS.CellULL1ThroughputLoad
2032	VS.DLPRBUsed
2033	VS.ULPRBUsed
2035	VS.DLPRBUsagePerTypeService
2037	VS.DLPRBAllocated
2038	VS.ULPRBAllocated
2039	VS.DLPRBUsedPerTypeService
2040	VS.ULPRBUsedPerTypeService
2041	VS.DLResidualMacBLERWithSemiPersistentScheduling
2042	VS.DLInitialMacBLERWithSemiPersistentScheduling
2043	VS.DLDataVolumeWithSemiPersistentScheduling
2044	VS.DLPRBUsedWithSemiPersistentScheduling
2045	VS.ULResidualMacBLERWithSemiPersistentScheduling
2046	VS.ULInitialMacBLERWithSemiPersistentScheduling
2047	VS.ULDataVolumeWithSemiPersistentScheduling
2048	VS.ULPRBUsedWithSemiPersistentScheduling
2056	VS.DLPDSCHResourceInefficiencyDueToLackPDCCHResource
2057	VS.ULPUSCHResourceInefficiencyDueToLackPDCCHResource
2058	VS.PUCCHSoundingReferenceSymbolConfigurationReject
2059	VS.PUCCHSoundingReferenceSymbolConfigurationSuccess
2060	VS.TTIUsageForPUSCHPerPRBGroup
2061	VS.DLeMbmsTotalPRBUsage
2062	VS.CellDLThroughput
2063	VS.DLEICICPercentagePRBUsage
2064	VS.CellBorderDLL1Throughput
2101	VS.NonGBRERABRlcThroughputDl
2102	VS.NonGBRERABRlcThroughputUl
2105	VS.DLRlcPduKbytes
2106	VS.ULRlcPduKbytes
2112	VS.GBRERABSatisfied
2113	VS.DLRlcPduSent
2114	VS.ULRlcPduReceived
2115	VS.DLRlcPduRetransmitted
12116	VS.TotalCountOfULTransportBlocks
12117	VS.TotalCountOfErrorULTransportBlocks
12118	VS.TotalCountOfDLTransportBlocks
12119	VS.TotalCountOfErrorDLTransportBlocks
12120	VS.NonGBRERABRlcThroughputDlLoad
12121	VS.NonGBRERABRlcThroughputUlLoad
12124	VS.DLRlcBurstTime
12125	VS.DLRlcLastTTITime
12126	VS.DLRlcBurstSize
12127	VS.DLRlcPduSizeInLastTTI
12200	VS.ENodeBTechnology
12201	VS.CounterReportingInformation
12202	VS.ReportGroupCustomerSelection
12303	RRC.ConnEstabFail.Sum
12304	RRC.ConnEstabFail
12305	VS.RadioLinkFailureSum
12306	VS.RadioLinkFailure
12307	RRC.ConnReEstabAtt.Sum
12308	RRC.ConnReEstabSucc
12309	RRC.ConnReEstabFail.Sum
12310	RRC.ConnReEstabFail
12311	RRC.ConnEstabAtt
12312	VS.RrcConnectionReleaseDueToMMEOverload
12314	VS.RrcConnectionReleaseDueToInabilityToPreempt
12315	VS.RrcConnectionRequestWithoutRepetition
12320	RRC.ConnEstabSucc
12321	VS.RrcConnectionSetupWithoutRepetition
12401	S1SIG.ConnEstabAtt
12402	VS.FirstDLNasTransport
12403	VS.UEContextSetupRequest
12405	VS.S1ConnectionEstablishmentFailure
12501	VS.InitialContextSetupSuccess
12502	VS.InitialContextSetupFailedSum
12503	VS.InitialContextSetupFailed
12504	VS.UEContextReleaseRequestSum
12505	VS.UEContextReleaseRequest
12506	VS.UEContextReleaseCommandSum
12507	VS.UEContextReleaseCommand
12508	VS.LocalUEContextRelease
12509	VS.LocalUEContextReleaseSum
12510	VS.UEContextModificationAttempt
12511	VS.UEContextModificationSuccess
12512	VS.UEContextModificationFailure
12513	VS.InitialContextSetupResponse
12603	VS.ERABSetupFailed
12604	SAEB.EstabInitAttNbr
12605	SAEB.EstabInitSuccNbr
12606	SAEB.EstabAddAttNbr
12607	SAEB.EstabAddSuccNbr
12608	VS.NormalERABRelease
12609	VS.AbnormalERABReleasePerQCI
12610	VS.IncomingERABToBeSetupOnIntraLteHO
12611	VS.IncomingERABSetupOnIntraLteHO
12612	SAEB.ModQoSAttNbr

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Counter Summary (Common / FDD / TDD) [cont.]

12613	SAEB.ModQoSSuccNbr
12614	SAEB.ModQoSFailNbr
12615	VS.ERABReleaseIndicationERABReleasedPerQCI
12616	VS.ERABReleaseIndicationERABReleasedPerCause
12617	VS.ERABReleaseCommandERABRequestedToBeReleasedPerQCI
12618	VS.ERABReleaseCommandERABRequestedToBeReleasedPerCaus
12619	VS.ERABReleaseResponseERABReleaseSuccess
12621	VS.IncomingERABToBeSetupOnIRATHO
12622	VS.IncomingERABSetupOnIRATHO
12630	VS.ERABReleasedDueToReactiveLoadControl
12631	VS.ERABSetupAttemptOverSPS
12632	VS.ERABSetupSuccessOverSPS
12633	VS.ERABSetupFailureOverSPS
12634	VS.ERABReleasedDueToRadioLinkFailurePerQCI
12635	VS.SAEBAbnormalRelActNbr
12701	VS.ReportedCellsNotSelected
12702	VS.IncomingIntraENodeBHOAttempt
12703	VS.IncomingIntraENodeBHOSuccess
12704	VS.IntraENodeBHOFailureSum
12705	VS.IntraENodeBHOFailure
12706	VS.OutgoingInterENodeBX2HOAttempt
12707	VS.OutgoingInterENodeBX2HOSuccess
12708	VS.OutgoingInterENodeBX2HOFailureSum
12709	VS.OutgoingInterENodeBX2HOFailure
12710	VS.IncomingInterENodeBX2HOAttempt
12711	VS.IncomingInterENodeBX2HOSuccess
12712	VS.IncomingInterENodeBX2HOFailureSum
12713	VS.IncomingInterENodeBX2HOFailure
12714	VS.NonOptimizedRedirectionToHRPDViaEventA2
12715	VS.RedirectionToUtraFdd
12716	VS.RedirectionToGeran
12717	VS.IntraCellHOAttempt
12718	VS.IntraCellHOSuccess
12719	VS.IntraCellHORekeyingFailure
12720	VS.OutgoingInterENodeBS1HOAttempt
12721	VS.OutgoingInterENodeBS1HOSuccess
12722	VS.OutgoingInterENodeBS1HOFailureSum
12723	VS.OutgoingInterENodeBS1HOFailure
12724	VS.IncomingInterENodeBS1HOAttempt
12725	VS.IncomingInterENodeBS1HOSuccess
12726	VS.IncomingInterENodeBS1HOFailureSum
12727	VS.IncomingInterENodeBS1HOFailure
12732	VS.OutgoingInterENodeBX2HOAbortSum
12733	VS.OutgoingInterENodeBX2HOAbort
12734	VS.IncomingInterENodeBX2HOAbortSum
12735	VS.IncomingInterENodeBX2HOAbort
12736	VS.OutgoingInterENodeBS1HOAbortSum
12737	VS.OutgoingInterENodeBS1HOAbort
12738	VS.IntraENodeBHOAbort
12739	VS.IntraCellHOKenodeBRefreshFailure
12742	VS.IntraENodeBHOAbortSum
12743	VS.IncomingInterENodeBS1HOAbortSum
12744	VS.IncomingInterENodeBS1HOAbort
12745	VS.OutgoingIntraENodeBHOAttempt
12746	HO.IntraEnbOutSucc.Sum
12747	VS.RedirectionToInterFrequencySameFrameStructure
12748	VS.RedirectionToUtraTdd
12755	VS.OutgoingPSHOToUtraTddAttempt
12756	VS.OutgoingPSHOToUtraTddSuccess
12757	VS.OutgoingPSHOToUtraTddFailureSum
12758	VS.OutgoingPSHOToUtraTddFailure
12759	VS.OutgoingPSHOToUtraTddAbortSum
12760	VS.OutgoingPSHOToUtraTddAbort
12761	VS.EnhancedNonOptimizedRedirectionToHRPD
12762	VS.CCOToGeranAttempt
12763	VS.CCOToGeranSuccess
12764	VS.CCOToGeranFailureSum
12765	VS.CCOToGeranFailure
12766	HO.IntraEnbOutAtt.Sum
12767	VS.OutgoingInterENodeBX2HOPreparationSuccess
12768	VS.IncomingInterENodeBX2HOPreparationSuccess
12769	VS.OutgoingInterENodeBS1HOPreparationSuccess
12770	VS.IncomingInterENodeBS1HOPreparationSuccess
12771	VS.OutgoingGapAssistedHOAttempt
12772	HO.InterFreqMeasGapOutSucc
12773	VS.OutgoingGapAssistedHOFailureSum
12774	VS.OutgoingGapAssistedHOAbortSum
12775	VS.X2RLFIndicationUnpreparedCell
12776	VS.OutgoingIntraFrequencyHOFailure
12777	VS.OutgoingIntraFrequencyHOMobilityEvent
12778	HO.IntraFreqOutFail
12779	VS.OutgoingIntraFrequencyHOMobilityEventPerRelation
12780	VS.OutgoingCsFallbackPSHOToUtraFddAttempt
12781	VS.OutgoingCsFallbackPSHOToUtraFddSuccess
12782	VS.OutgoingCsFallbackPSHOToUtraFddFailureSum
12783	VS.OutgoingCsFallbackPSHOToUtraFddAbortSum
12784	VS.CsFallbackCCOToGeranAttempt
12785	VS.CsFallbackCCOToGeranSuccess
12786	VS.CsFallbackCCOToGeranFailureSum
12788	VS.OutgoingPSHOToUtraFddSuccess
12789	VS.OutgoingPSHOToUtraFddFailureSum
12790	VS.OutgoingPSHOToUtraFddFailure
12791	VS.OutgoingPSHOToUtraFddAbortSum
12792	VS.OutgoingPSHOToUtraFddAbort
12793	VS.OutgoingPSHOToUtraFddPreparationSuccess
12794	VS.EvolvedMultiCarrierTrafficAllocationTrigger
12797	VS.IncomingPSHOFromUtranAttempt
12798	VS.IncomingPSHOFromUtranPreparationSuccess
12799	VS.IncomingPSHOFromUtranSuccess
12801	VS.IncomingPSHOFromUtranFailureSum
12802	VS.IncomingIntraENodeBHOAttemptScreened
12803	VS.IncomingIntraENodeBHOSuccessScreened
12804	VS.IncomingPSHOFromUtranFailure
12805	VS.IncomingPSHOFromUtranAbortSum
12806	VS.OutgoingInterENodeBX2HOAttemptScreened
12807	VS.OutgoingInterENodeBX2HOSuccessScreened
12808	VS.IncomingPSHOFromUtranAbort
12810	VS.IncomingInterENodeBX2HOAttemptScreened

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Counter Summary (Common / FDD / TDD) [cont.]

12810	VS.IncomingInterENodeBX2HOAttemptScreened
12811	VS.IncomingInterENodeBX2HOSuccessScreened
12812	VS.OutgoingEmergencyCsFallbackPSHOToUtraFddAttempt
12813	VS.OutgoingEmergencyCsFallbackPSHOToUtraFddSuccess
12814	VS.OutgoingEmergencyCsFallbackPSHOToUtraFddFailureSum
12815	VS.OutgoingEmergencyCsFallbackPSHOToUtraFddAbortSum
12816	VS.EmergencyCsFallbackCCOToGeranAttempt
12817	VS.EmergencyCsFallbackCCOToGeranSuccess
12818	VS.EmergencyCsFallbackCCOToGeranFailureSum
12819	VS.OutgoingPSHOToUtraTddFailurePerHOReason
12820	VS.OutgoingInterENodeBS1HOAttemptScreened
12821	VS.OutgoingInterENodeBS1HOSuccessScreened
12822	VS.OutgoingIntraENodeBInterPlmnHOAttempt
12823	VS.OutgoingInterENodeBInterPlmnX2HOAttempt
12824	VS.IncomingInterENodeBS1HOAttemptScreened
12825	VS.IncomingInterENodeBS1HOSuccessScreened
12826	VS.OutgoingSrvccToUtraFddAttempt
12827	VS.OutgoingSrvccToUtraFddSuccess
12828	VS.OutgoingSrvccToUtraFddFailureSum
12829	VS.OutgoingSrvccToUtraFddFailure
12830	VS.OutgoingSrvccToUtraFddAbortSum
12831	VS.OutgoingSrvccToUtraFddAbort
12832	VS.OutgoingInterENodeBX2HOAbortScreenedSum
12833	VS.OutgoingInterENodeBInterPlmnS1HOAttempt
12834	VS.IncomingInterENodeBX2HOAbortScreenedSum
12835	VS.OutgoingSrvccToUtraFddFailurePerHOReason
12836	VS.OutgoingInterENodeBS1HOAbortScreenedSum
12837	VS.OutgoingIntraENodeBInterPlmnHOSuccess
12838	VS.OutgoingInterENodeBInterPlmnX2HOSuccess
12839	VS.OutgoingInterENodeBInterPlmnS1HOSuccess
12840	VS.OutgoingSrvccToUtraTddFailurePerHOReason
12841	VS.OutgoingPSHOToUtraTddPreparationSuccess
12842	VS.IntraENodeBHOAbortScreenedSum
12843	VS.IncomingInterENodeBS1HOAbortScreenedSum
12845	VS.OutgoingIntraENodeBHOAttemptScreened
12846	VS.OutgoingIntraENodeBHOSuccessScreened
12851	VS.RedirectionTo1xRtt
12853	VS.CsFallbackRequest
12858	VS.EnhancedRedirectionToUtraFdd
12859	VS.EnhancedRedirectionToGeran
12860	VS.OutgoingSrvccToUtraTddAttempt
12861	VS.OutgoingSrvccToUtraTddSuccess
12862	VS.OutgoingSrvccToUtraTddFailureSum
12863	VS.OutgoingSrvccToUtraTddFailure
12864	VS.OutgoingSrvccToUtraTddAbortSum
12865	VS.OutgoingSrvccToUtraTddAbort
12866	VS.IntraENodeBHOPreparationSuccessScreened
12867	VS.OutgoingInterENodeBX2HOPreparationSuccessScreened
12868	VS.IncomingInterENodeBX2HOPreparationSuccessScreened
12869	VS.OutgoingInterENodeBS1HOPreparationSuccessScreened
12870	VS.IncomingInterENodeBS1HOPreparationSuccessScreened
12875	VS.OutgoingEmergencyCsFallbackPSHOToUtraTddSuccess
12876	VS.OutgoingEmergencyCsFallbackPSHOToUtraTddFailureSum
12877	VS.OutgoingEmergencyCsFallbackPSHOToUtraTddAbortSum
12878	VS.OutgoingEmergencyCsFallbackPSHOToUtraTddAttempt
12889	VS.OutgoingPSHOToUtraFddFailurePerHOReason
12890	VS.OutgoingInterENodeBS1HOAbortPerHOReason
12891	VS.OffLoadingSuccess
12892	VS.OffLoadingFailure
12893	VS.OutgoingInterENodeBX2HOAbortPerHOReason
12894	VS.EnhancedRedirectionToUtraTdd
12895	VS.OutgoingCsFallbackPSHOToUtraTddAttempt
12896	VS.OutgoingCsFallbackPSHOToUtraTddSuccess
12897	VS.OutgoingCsFallbackPSHOToUtraTddFailureSum
12898	VS.OutgoingCsFallbackPSHOToUtraTddAbortSum
12909	VS.X2ReceivedThroughput
12910	VS.X2ReceivedPackets
12911	VS.X2SentThroughput
12912	VS.X2SentPackets
13001	VS.NbUeScheduledPerDLTTI
13002	VS.NbUeScheduledPerULTTI
13003	VS.ULGrant
13004	VS.ULPairedGrant
13005	VS.DLGrant
13006	VS.ULPHnormalized
13007	VS.DLMimoEligibilityDecision
13008	VS.ContentionBasedRandomAccessPreamble
13009	VS.ContentionFreeRandomAccessPreamble
13010	VS.ContentionBasedRandomAccessResponse
13011	VS.ContentionFreeRandomAccessResponse
13012	VS.ContentionResolution
13013	VS.Layer0TxDivWBCqiReported
13014	VS.Layer0MimoWBCqiReported
13015	VS.Layer1WBCqiReported
13017	VS.TransmissionMode8UsersDistribution
13019	VS.RRHAreaChanges
13109	VS.S1DLThroughput
13110	VS.S1DLPackets
13111	VS.S1ULThroughput
13112	VS.S1ULPackets
13201	RRC.Conn
13204	VS.NbVoIPBearersPerCellPerPlmn
13205	VS.NbGBRBearersPerCellPerPlmn
13206	VS.NbNonGBRBearersPerCellPerPlmn
13207	VS.NbBearersPerCell
13208	VS.NbVoIPBearersPerENodeB
13209	VS.NbGBRBearersPerENodeB
13210	VS.NbNonGBRBearersPerENodeB
13211	VS.NbBearersPerENodeB
13212	VS.NbActiveUsersPerENodeB
13213	VS.DLPRBsPoolOverloadScreened
13214	VS.ULPRBsPoolOverloadScreened
13215	VS.NbActiveUEInULPerTypeService
13216	VS.NbActiveUEInDLPerTypeService
13217	VS.NbBearersPerCellPerQCIPerPlmn
13218	VS.NbBearersPerENodeBPerQCI
13219	VS.NpuCpu1UtilizationHistogram
13220	VS.ENodeBControlCpuUtilizationHistogram

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Counter Summary (Common / FDD / TDD) [cont.]

13221	SAEB.ModQoSSuccNbr
13222	SAEB.ModQoSFailNbr
13223	VS.ERABReleaseIndicationERABReleasedPerQCI
13224	VS.ERABReleaseIndicationERABReleasedPerCause
13225	VS.ERABReleaseCommandERABRequestedToBeReleasedPerQCI
13226	VS.ERABReleaseCommandERABRequestedToBeReleasedPerCause
13227	VS.ERABReleaseResponseERABReleaseSuccess
13228	VS.IncomingERABToBeSetupOnIRATHO
13229	VS.IncomingERABSetupOnIRATHO
13230	VS.ERABReleasedDueToReactiveLoadControl
13231	VS.ERABSetupAttemptOverSPS
13232	VS.ERABSetupSuccessOverSPS
13233	VS.ERABSetupFailureOverSPS
13234	VS.ERABReleasedDueToRadioLinkFailurePerQCI
13235	VS.SAEBAbnormalRelActNbr
13236	VS.ReportedCellsNotSelected
13237	VS.IncomingIntraENodeBHOAttempt
13238	VS.IncomingIntraENodeBHOSuccess
13239	VS.IntraENodeBHOFailureSum
13240	VS.IntraENodeBHOFailure
13241	VS.OutgoingInterENodeBX2HOAttempt
13242	VS.OutgoingInterENodeBX2HOSuccess
13243	VS.OutgoingInterENodeBX2HOFailureSum
13244	VS.OutgoingInterENodeBX2HOFailure
13245	VS.IncomingInterENodeBX2HOAttempt
13246	VS.IncomingInterENodeBX2HOSuccess
13249	VS.IncomingInterENodeBX2HOFailureSum
13301	VS.IncomingInterENodeBX2HOFailure
13302	VS.NonOptimizedRedirectionToHRPDViaEventA2
13303	VS.RedirectionToUtraFdd
13304	VS.RedirectionToGeran
13305	VS.IntraCellHOAttempt
13306	VS.IntraCellHOSuccess
13307	VS.IntraCellHORekeyingFailure
13308	VS.OutgoingInterENodeBS1HOAttempt
13309	VS.OutgoingInterENodeBS1HOSuccess
13310	VS.OutgoingInterENodeBS1HOFailureSum
13311	VS.OutgoingInterENodeBS1HOFailure
13312	VS.IncomingInterENodeBS1HOAttempt
13313	VS.IncomingInterENodeBS1HOSuccess
13314	VS.IncomingInterENodeBS1HOFailureSum
13315	VS.IncomingInterENodeBS1HOFailure
13316	VS.OutgoingInterENodeBX2HOAbortSum
13317	VS.OutgoingInterENodeBX2HOAbort
13318	VS.IncomingInterENodeBX2HOAbortSum
13319	VS.IncomingInterENodeBX2HOAbort
13320	VS.OutgoingInterENodeBS1HOAbortSum
13321	VS.OutgoingInterENodeBS1HOAbort
13326	VS.IntraENodeBHOAbort
13327	VS.IntraCellHOKenodeBRefreshFailure
13328	VS.IntraENodeBHOAbortSum
13329	VS.IncomingInterENodeBS1HOAbortSum
13330	VS.IncomingInterENodeBS1HOAbort
13331	VS.OutgoingIntraENodeBHOAttempt
13332	HO.IntraEnbOutSucc.Sum
13333	VS.RedirectionToInterFrequencySameFrameStructure
13334	VS.RedirectionToUtraTdd
13335	VS.OutgoingPSHOToUtraTddAttempt
13336	VS.OutgoingPSHOToUtraTddSuccess
13337	VS.OutgoingPSHOToUtraTddFailureSum
13401	VS.OutgoingPSHOToUtraTddFailure
13402	VS.OutgoingPSHOToUtraTddAbortSum
13403	VS.OutgoingPSHOToUtraTddAbort
13404	VS.EnhancedNonOptimizedRedirectionToHRPD
13405	VS.CCOToGeranAttempt
13406	VS.CCOToGeranSuccess
13407	VS.CCOToGeranFailureSum
13408	VS.CCOToGeranFailure
13409	HO.IntraEnbOutAtt.Sum
13410	VS.OutgoingInterENodeBX2HOPreparationSuccess
13501	VS.IncomingInterENodeBX2HOPreparationSuccess
13502	VS.OutgoingInterENodeBS1HOPreparationSuccess
13601	VS.IncomingInterENodeBS1HOPreparationSuccess
13602	VS.OutgoingGapAssistedHOAttempt
13603	VS.S1SctpInOctets
13604	VS.S1SctpInPackets
13605	VS.S1SctpOutOctets
13606	VS.S1SctpOutPackets
13607	VS.X2SctpInOctets
13608	VS.X2SctpInPackets
13609	VS.X2SctpOutOctets
13610	VS.X2SctpOutPackets
13701	VS.RSSIMeasurement
13702	VS.RFMTxPowerMeasurement
13703	VS.RFMProcessorOccupancy
13704	VS.CellTxPowerMeasurement
13801	VS.CACRequest
13802	VS.CACFailure
13803	VS.VlanTransportCACFailureOnS1u
13804	VS.PortTransportCACFailureOnS1u
13805	VS.VlanTransportCACFailureForEmergencyCallOnS1u
13806	VS.PortTransportCACFailureForEmergencyCallOnS1u
13807	VS.CACRequestForPlmn
13808	VS.CACFailureForPlmn
13901	VS.OtdoaInformationRequest
13902	VS.OtdoaInformationFailure
13903	VS.OtdoaInformationResponse
13904	VS.EcidMeasurementInitiationRequest
13905	VS.EcidMeasurementInitiationFailure
13906	VS.EcidMeasurementInitiationResponse
14101	VS.S1ErrorIndicationByENodeB
14102	VS.S1ErrorIndicationByMME
14201	VS.DRBPdcpSduKbytesDL
14202	VS.DRBPdcpSduKbytesUL
14203	VS.DRBPdcpSduBitRateDL
14204	DRB.PdcpSduBitrateUl
14205	VS.SRBPdcpSduKbytesDL

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Counter Summary (Common / FDD / TDD) [cont.]

14206 VS.SRBPdcpSduKbytesUL
14301 VS.M1GtpPayloadKbytesReceived
14302 VS.M1SyncSequencesReceivedTooEarly
14303 VS.M1SyncSequencesReceivedTooLate
14304 VS.M1SyncSequencesDelay
14305 VS.MbmsUserPacketsExpectedBySyncLayer
14306 VS.MbmsUserPacketsReceivedBySyncLayer
14307 VS.MbmsUserPacketsReceivedByRlc
14308 VS.MbmsUserPacketsDroppedByRlcUponOverflow
14309 VS.M1GtpPayloadKbytesReceivedPerENodeB
14310 VS.M1SyncSequencesReceivedTooEarlyPerSessionPerArea
14311 VS.M1SyncSequencesReceivedTooLatePerSessionPerArea
14312 VS.M1SyncSequencesDelayPerSessionPerArea
14313 VS.MbmsUserPacketsExpectedBySyncLayerPerSessionPerArea
14314 VS.MbmsUserPacketsReceivedBySyncLayerPerSessionPerArea
14315 VS.MbmsUserPacketsReceivedByRlcPerSessionPerArea
14316 VS.MbmsUserPacketsDroppedByRlcUponOverflowPerSessionPerArea
14401 VS.TransmissionModeSwitchRequest
14402 VS.TransmissionModeSwitchSuccess
14501 VS.MbmsSessionStartRequest
14502 VS.MbmsSessionStartResponse
14503 VS.MbmsSessionStartFailure
14504 VS.MbmsSessionUpdateRequest
14505 VS.MbmsSessionUpdateResponse
14506 VS.MbmsSessionUpdateFailure
14507 VS.MbmsSessionStopRequest
14508 VS.MbmsSessionStopResponse
14509 VS.M3Reset
14510 VS.NbActiveMbmsSessions
14511 VS.MbmsSchedulingInformation
14512 VS.MbsfnSubframeAllocationEventsPerCell
14513 VS.IGMPv3MembershipQuery
14514 VS.IGMPv3MembershipReportState
14515 VS.IGMPv3MembershipReportJoin
14516 VS.IGMPv3MembershipReportLeave
14517 VS.MLDv2MulticastListenerQuery
14518 VS.MLDv2MulticastListenerReportState
14519 VS.MLDv2MulticastListenerReportJoin
14520 VS.MLDv2MulticastListenerReportLeave

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End of module
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